

Structures in C

Algorithmic Notation & C Quick Reference Card

1. Declaration

Algorithm	C equivalent
Structure NomStructure	typedef struct {
champ1 : type1	type1 champ1;
champ2 : type2	type2 champ2;
Fin Structure	} NomStructure;

2. Variables

Algorithm	C equivalent
Var x : Structure X	X x;
Var arr : Tableau[50] de X	X arr[50];
Var x : entier	int x;
Var x : réel	float x;
Var x : chaîne	char x[50];
Var x : caractère	char x;
Var x : booléen	int x; // 0 or 1

3. Accessing Fields

Algorithm	C equivalent
x.champ	x.field
arr[i].champ	arr[i].field
x.sousstruct.champ	x.substruct.field
arr[i].sousstruct.champ	arr[i].substruct.field

4. Operations

Algorithm	C equivalent
x ← valeur	x = valeur;
Lire(x.champ)	scanf("%d", &x.field);
Lire(x.nom)	scanf("%s", x.name); // no &
Écrire(x.champ)	printf("%d\n", x.field);

5. Loops

Algorithm	C equivalent
Pour i ← 0 à n-1 Faire	for (int i = 0; i < n; i++) {
...	...

Fin Pour	}
TantQue cond Faire	while (condition) {
...	...
Fin TantQue	}

6. Conditions

Algorithm	C equivalent
Si condition Alors	if (condition) {
...	...
Sinon	} else {
...	...
Fin Si	}

7. scanf Rules — When to use &

Type	scanf example
char name[50] → NO &	scanf("%s", s.name);
int age → YES &	scanf("%d", &s.age);
float price → YES &	scanf("%f", &s.price);
char grade → YES &	scanf(" %c", &s.grade);

8. Loop Start Rules

Loop type	Start from
Input loop	0 – fill every slot
Print loop	0 – show every element
Find max/min loop	1 – index 0 is already the default

* Never use spaces in folder/file names when coding — use _ or - instead.